Teacher sheet lesson one

Introduction

This unit will be organised around an edStudio presentation that will contain the majority of the student resources. All of the investigations will be listed in the find and prepare sections of the lesson plans. Each page of the edStudio will be a new lesson and the students can individually work through the activities and submit their produced work electronically back onto the pages via the digital drop boxes.

The edStudio presented in this unit will need to be personalised for each individual class. You will need to make a copy of the edStudio and then alter the permissions and/or modify the content to suit specific classes.

It is possible to teach the lessons without the edStudio. If you do not wish to use it, a full list of the resources used has been included in the lesson plans. Student instructions for activities, that appear as pop-up boxes in the studio, are included on each teacher sheet as student information.

Introduction — The structure of the Earth (website)

Student information:

Welcome to the unit. Today we will be looking at a slideshow that revises what you learned about rocks in Year 8 and introduces the structure of the Earth. You will then be working in groups to build up a model of the Earth. Please follow the instructions on this page to work your way through lesson one. First watch the slideshow *Structure of the Earth* (http://media2.teachertube.com/files/articles/3933.ppt#256,1,Unit 17 STRUCTURE OF THE EARTH)

Afterwards, your teacher will lead a discussion on what you have viewed.

Task 1 — Introductory reading

Student information:

Open the website *Structure of the Earth* (http://scign.jpl.nasa.gov/learn/plate1.htm) and read the information about the layers of the Earth. Do not navigate away from this page. When you have finished reading, create a page of notes of the relevant summary points, then create a labelled scale drawing of the four layers of the Earth. You may complete this task in your science journal or you may create an electronic document and save it in a new folder for this unit.

Teacher information:

The *Structure of the Earth* website contains a few interconnecting pages. Ensure the students only summarise the information on the page that opens from the link.

Task 2 — Make a model of the Earth

Student information:

Now you have read about the structure of the Earth you will construct a model. Collect your materials then follow the supplied instructions. Make a recording of your group building the model and submit it to your teacher. If you are unsure of how to record, read through the section ‘How to prepare a presentation’ below. Once you have finished, view another group’s recordings and models.

Teacher information:

Student instructions are provided at the end of this document. For this activity the students will need the handouts ‘Make a model of the Earth’ and ‘How to prepare a presentation’.

Task 3 — From clay balls to the centre of the Earth

Student information:

In task two you constructed a model of the Earth from a description based on accepted scientific theory. How do we know what the Earth is composed of if we can never see all the layers? Use the activity *From clay balls to the structure of the Earth* to gain some ideas about how scientists are using experiments to discover what the inside of the Earth is made of.

Teacher information:

Refer to the website *From clay balls to the structure of the Earth* (http://geophysics.esci.keele.ac.uk/earthlearningidea/PDF/74_Clay_balls.pdf). A student activity sheet of the same title is included at the end of this document. You will need to personalise this activity to suit your class.

Culminating activity — Discussion questions

Student information:

Think about the two activities that you have completed and then read through the discussion questions below and write your answers in your science journal. Consider other students’ answers and reflect on whether you agree or disagree.

Discussion questions:

1. What did you predict the centre of the two plasticine balls was composed of?
2. Do you need to cut open the balls to make this prediction?
3. Would you need to open the balls in order to draw a definitive conclusion?
4. What is the difference between a prediction and a conclusion?
5. Have scientists made predictions about the Earth’s composition or drawn conclusions?
6. Will scientists ever be able to draw conclusions about the composition of the Earth?

It is important for this resource that misconceptions are not created by the students engaging with the activity. Explain to the students that the activities they complete on the clay balls are not mini-experiments that mirror current scientific investigation. They are, however, investigating that it is possible to form hypotheses based on experimental evidence, which is how scientists are currently developing a concept of the structure of the Earth.

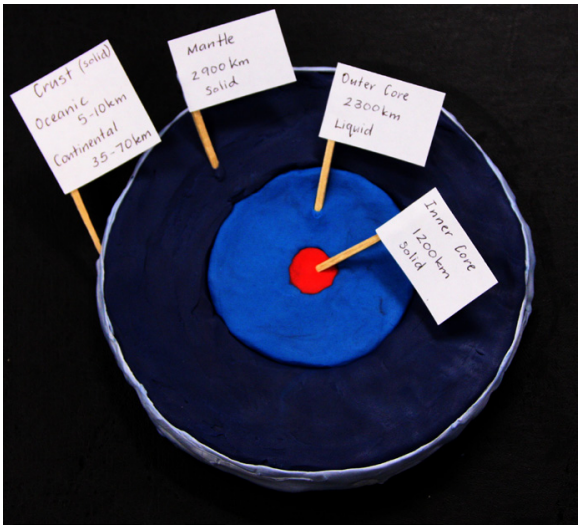
Make a model of the Earth

Materials:

- Four colours of modelling clay
- Piece of paper for labels
- Four toothpicks
- Plastic knife
- Metric ruler

Method:

1. You first need to be in groups of two or three and then collect your materials from the front of the room. Remember to organise the recording of your activity now.
2. Each group will be making one labelled model of a slice of the Earth. Use toothpicks and paper to make the labels. Below is an image of a finished product.



Schoch, Catherine © The State of Queensland

3. This will be a scale model so the group will need to work out how thick each layer should be. You will be making a 1 cm:1000 km model. Look up the thickness of each layer in your notes from task one. You will need to include the inner core, outer core, mantle and continental crust as four separate layers (the oceanic crust will be impossibly thin on this model so use the continental crust thickness). Remember to record yourselves working out this information.
4. Once you have worked out the measurements for your model you may commence building it. Use a different colour of plasticine for each layer. You need to be careful that each layer is the correct thickness. One member of your team needs to narrate your procedure while you record building your model.
5. Once the model is completed you may make some little flags from toothpicks and paper. Label each layer with its name, size and composition. Don't forget to keep recording.
6. Now hold up your model for the camera and give a short presentation on your finished model, including how and why you made construction decisions.
7. Your group is now finished this activity. Upload your recording to a computer and make some final editing adjustments before you submit the final presentation.

How to prepare a presentation

You will be completing several experiments during the course of your study this year. As part of these lessons you will be required to record your participation in and reflection on each one. We will be recording in an electronic format and keeping digital copies of these records.

So, how do I do that?

Good question. You will be provided with some form of recording equipment at the beginning of the lesson. This may consist of a camera, a video recorder or a digital voice recorder. You will use these items to make a recording of the experiment as your group is completing the activity.

Remember, you cannot go back and record an experiment once your group has completed it.

Things to consider

Your presentation on your experiment will still need to have the elements of a normal experiment 'write up'. So you will need to state in your presentation the following elements:

- The aim of the experiment — what is your group trying to discover?
- Your group's initial hypothesis — what does your group think you are going to observe?
- How your group is intending to investigate this hypothesis — this is an initial discussion of your method.
- Images of your equipment during and after set up — name the equipment and say why it is the most suited to the task (e.g. because we are only using small measures of materials we will use test tubes instead of beakers).
- A commentary of your experiment method including any safety considerations your group has recognised.
- A commentary and any images of your observations — what did your group see during the experiment?
- Your concluding statements, and a discussion of these questions:
  - Why did you see your observations?
  - Are they what you expected to see?
  - Did you have observations that may have been caused by errors in your method?
- Your group's reflection on the experiment and an analysis of your level of success or failure:
  - How did your group manage the experiment?
  - How many errors did you think you made?
  - How did your results compare with your hypothesis?

In a written presentation you would normally write out descriptions of these points, but for an electronic presentation your group will need to speak about these aloud as the group is working through the experiment.

It may be a good idea to nominate someone to act as the 'presenter' for the recording. They will be responsible for the main commentary and for asking questions of the other group members.

Once the experiment is finished and cleaned away you can download your presentation and give it some finishing touches. If you are taking photos and recording your voices, then a slideshow may be the best option for your format.

If you have a video recording, then some general editing and additional audio via a voice recorder may be a good idea.

From clay balls to the structure of the Earth

You made a scale model of the Earth for task two. How do we know that these predictions about the Earth's composition are true? The Earth is very large and no one has been able to drill all the way to the core. This is impossible.

For task three you will be looking at how scientists have come to formulate predictions on the composition of the Earth's layers. Follow the steps below and conduct some tests on two objects to see if you can predict what the objects are composed of.

Step one: Form into groups of two or three. Collect your materials from the front of the room.

What are the two objects?

Do they look the same?

Do you think they are the same? Explain your answer.

What do you think the inside of the objects look like? Note: You are not to cut the objects open during this activity.

Make a list of the tests you could carry out on the two balls and then wait for the teacher to lead you in a discussion of your classmates' ideas.

Now take your two balls and complete some of the tests on them. Make a record of your observations for each investigation.